CA6126 – Reinforcement Learning

Sequential Multi-Target Delivery under Partial Observability

Technical Report

Project Title: Sequential Multi-Target Delivery under Partial Observability:
A POMDP Approach with Recurrent Policy Gradient Methods

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1 Introduction

We study a reinforcement learning problem in which a delivery robot must visit K landmarks on an $N \times N$ grid in a *hidden* order that is randomised at every episode. The agent receives only binary feedback (correct / wrong) after each visit attempt, and must therefore *infer* the ordering from its interaction history. This makes the task a Partially Observable Markov Decision Process (POMDP), where the optimal policy is inherently non-Markov in the observation and requires memory.

The problem is inspired by real-world scenarios such as multi-stop delivery routing where the priority of stops is unknown to the driver and can only be discovered on arrival. It is not a Travelling Salesman Problem: the order is fixed but unknown, and the agent must trade off *information gain* against *travel cost.*

We implement the environment as a Gymnasium-compatible Python module and train agents using the Stable Baselines 3 ecosystem. Our key contribution is a **systematic investigation of why standard RL algorithms fail on this POMDP** and the design choices needed to enable successful learning: reward shaping, penalty tuning, and problem scaling.

2 Problem Formulation

2.1 POMDP Specification

We formalise the problem as a POMDP tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, O, \gamma)$:

Hidden state s_t.

$$s_t = \big(p_t,\ \{\ell_i\}_{i=1}^{K},\ \pi,\ k_t,\ t\big)$$

where $p_t \in \{0, \ldots, N-1\}^2$ is the robot position, ℓ_i are landmark positions, $\pi \in S_K$ is the hidden visiting permutation, k_t is the number of correct visits so far, and t is the step counter.

Observation o_t. The agent observes:

- A 5×5 ego-centric view (one-hot encoding over {empty, $\text{landmark}_1, \ldots, \text{landmark}_K$, wall}).
- A history vector $h_t \in \{-1, 0, +1\}^K$: -1 = tried & rejected, 0 = unknown, $+1$ = confirmed correct.
- Progress k_t/K and normalised remaining steps $(T-t)/T$.

Crucially, the permutation π is **never** revealed.

Action space. $\mathcal{A} = \{\text{Up, Down, Left, Right, Visit}\}$, $|\mathcal{A}| = 5$.

Event	**Reward**
Per-step cost (any action)	-0.01
Wall bump	-0.05
Visit on empty cell	-0.05
Correct landmark visit	$+1.00$
Wrong landmark visit	$-p$ (configurable)
Completing full sequence ($k = K$)	$+5.00$
Timeout ($t = T$)	-1.00

Table 1: Reward function design. The wrong-visit penalty p is a key hyperparameter investigated in this work (default $p = 0.5$).

Reward structure.

2.2 Why This Is Novel

- Standard gridworlds expose the goal; here the goal-ordering is hidden.
- Unlike TSP, the order is fixed but unknown—pure shortest-path heuristics fail.
- The optimal policy requires *sequential hypothesis testing*: the agent must actively explore to narrow down the $K!$ possible orderings.
- The history vector h_t is a sufficient statistic for belief tracking, but learning to *use* it is non-trivial.

Reward shaping. We implement a potential-based shaping term (Ng et al., 1999):

$$F(s, s') = \gamma\,\Phi(s') - \Phi(s), \qquad \Phi(s) = -\frac{d_{\min}(p, \text{plausible landmarks})}{2(N-1)}$$

where $d_{\min}$ is Manhattan distance to the nearest landmark with $h_i = 0$ (untried). This gives a continuous positive gradient toward landmarks without changing the optimal policy in theory.

3 Environment Implementation

The environment is implemented as a Gymnasium `Env` subclass (`env.py`) using the Stable Baselines 3 framework for all algorithm implementations. Key design decisions:

1. **Dict observation space:** Separate ego-view, history, progress, and time budget into distinct keys for SB3's `MultiInputPolicy`.
2. **Deterministic dynamics:** Stochasticity comes only from the hidden permutation and initial placement.
3. **Configurable difficulty:** Grid size (N), number of landmarks (K), view size, wrong-visit penalty (p), and reward shaping are all adjustable.
4. **Headless rendering:** Pygame-based renderer (`render.py`) produces RGB frames for video recording.

4 Algorithms

All algorithms are from the Stable Baselines 3 (SB3) ecosystem:

- **PPO:** Feed-forward `MultiInputPolicy` with $[256, 256]$ MLPs. 16 parallel envs.
- **Recurrent PPO (LSTM):** `MultiInputLstmPolicy` from `sb3-contrib` with 128-unit LSTM + $[128, 128]$ MLPs. The LSTM maintains internal state across timesteps, enabling belief tracking.
- **DQN:** $[256, 256]$ MLP with experience replay (200k buffer), ε-greedy exploration.
- **QR-DQN:** Distributional DQN with 51 quantiles.

5 Experimental Journey: From Failure to Success

This section documents our iterative debugging process, which is central to understanding how to solve POMDPs with deep RL.

5.1 Phase 1: Large Grid Baseline (12×12, $K{=}4$, $p{=}0.5$)

Setup. Our initial configuration followed standard practice: 12×12 grid, 4 landmarks ($4! = 24$ orderings), 5×5 ego-view, wrong-visit penalty $p = 0.5$, no reward shaping. We trained PPO, DQN, and RecurrentPPO for 5M steps each.

Algorithm	Reward	Success Rate	Wrong Visits	Ep. Length
Random baseline	-5.95 ± 0.94	0.0%	0.99	200.0
PPO (3 seeds)	-3.00 ± 0.00	0.0%	0.00	200.0
DQN (seed 0)	-3.97 ± 7.67	0.0%	1.73	200.0
RecurrentPPO	-3.00 ± 0.00	0.0%	0.00	200.0

Table 2: Phase 1: All algorithms fail on the large grid. 200 eval episodes.

Results.

Analysis: The "Never Visit" Local Optimum. Every algorithm converged to a policy that *never executes the Visit action*. The reward decomposition reveals why:

$$\underbrace{200 \times (-0.01)}_{\text{step cost}} + \underbrace{(-1.0)}_{\text{timeout}} = -3.0$$

Meanwhile, each exploratory visit attempt has **negative expected value**:

$$\mathbb{E}[\text{single visit}] = \frac{1}{4} \times (+1.0) + \frac{3}{4} \times (-0.5) = -0.125$$

The agent rationally concludes that *doing nothing* (-3.0) beats *trying* (expected < -3.0). This is a fundamental exploration problem: the large positive reward from completing the sequence ($+8.0$) is too unlikely to discover through random exploration.

Additional factors.

- **Low visibility:** 5×5 view covers only $25/144 = 17\%$ of the 12×12 grid. Landmarks are usually invisible.
- **High combinatorial complexity:** $4! = 24$ orderings require extensive exploration to disambiguate.
- DQN performs worst (-3.97, 1.73 wrong visits) because ε-greedy forces random Visit actions, incurring penalties.

5.2 Phase 2: Adding Reward Shaping (12×12, $K=4$, $p=0.5$, shaped)

Hypothesis. The agent cannot discover landmarks because it has no gradient signal pointing toward them. Potential-based reward shaping should guide navigation without changing the optimal policy.

Results. RecurrentPPO with shaping reached train reward -2.69 at 2M steps (vs. -3.03 without shaping). The improvement indicates the agent learned to **approach landmarks**, but eval reward remained -3.00 with 0% success—the best model saved by EvalCallback still corresponded to the "never visit" strategy because eval uses unmodified rewards.

Takeaway. Reward shaping helps navigation but is insufficient alone: the agent approaches landmarks but still does not dare to visit them.

5.3 Phase 3: Reducing Wrong-Visit Penalty (12×12, $K=4$, $p=0.1$, shaped)

Hypothesis. With $p = 0.1$, a single visit attempt has positive expected value:

$$\mathbb{E}[\text{single visit}] = \frac{1}{4} \times (+1.0) + \frac{3}{4} \times (-0.1) = +0.175 > 0$$

This should incentivise exploration.

Results. RecurrentPPO with shaping + low penalty reached −2.62 at 1.3M steps, slightly better than Phase 2 at the same stage. However, eval success rate remained 0%—the 12×12 grid is still too large for the agent to reliably find and visit landmarks.

Takeaway. Lowering penalty makes exploration rational in expectation, but spatial exploration on a large grid with limited vision remains the bottleneck.

5.4 Phase 4: Scaling Down (6×6, $K=3$, $p=0.1$, shaped) — Breakthrough

Hypothesis. The core POMDP challenge (inferring hidden order) is orthogonal to the spatial navigation challenge. By reducing the grid to 6×6 (view coverage: $25/36 = 69\%$), landmarks to 3 (only $3! = 6$ orderings), and episode length to 100 steps, we isolate the POMDP reasoning from spatial difficulty.

Agent	Reward (mean±std)	Success Rate	Best Episode
Random baseline (6×6, $K=3$)	~ -3.0	0.0%	—
RecurrentPPO seed 0	$+6.09 \pm 3.04$	**87.5%**	+7.91
RecurrentPPO seed 1	-2.00 ± 0.11	0.0%	−1.10
RecurrentPPO seed 2	$+4.79 \pm 4.59$	**78.5%**	+7.89

Table 3: Phase 4: Small grid results across 3 seeds. Seeds 0 and 2 solve the task; seed 1 fails.

Results (3 seeds, 2M steps each, 200 eval episodes per seed).

Why seed 1 failed: the exploration lottery. Seed 1 converged to the same "never visit" strategy (−2.0 reward) as the large-grid experiments. The root cause is **early exploration stochasticity**: the initial random weights and early trajectory samples determine whether the agent discovers a correct visit within the first ∼100k steps. Seeds 0 and 2 happened to stumble upon a correct visit early, providing a positive reward signal (+1.0) that the LSTM could latch onto and generalise. Seed 1 never received this "lucky" signal and remained stuck in the local optimum.

This is a well-known phenomenon in RL: **the same hyperparameters can yield vastly different outcomes depending on the random seed**. With 3 seeds, we observe a 67% success rate at the seed level (2 out of 3 seeds solve the task), suggesting the algorithm is effective but not fully robust. Strategies to improve seed robustness include:

- Higher entropy coefficient to encourage exploration.
- Curriculum learning: start with $K = 2$ landmarks and increase.
- Longer training with exploration bonuses (e.g., intrinsic curiosity).

Training dynamics (seed 0). The train reward trajectory tells a clear story of learning progression:

- **0–50k steps:** Random exploration, reward ~ -3.15.
- **50k–200k:** Agent learns navigation toward landmarks ($-3.15 \to -2.59$), driven by reward shaping.
- **200k–800k:** Agent begins attempting visits, reward improves through correct deliveries ($-2.59 \to -1.87$).
- **800k–1.5M:** LSTM learns to track history and plan multi-step visit sequences ($-1.87 \to -1.35$ train).
- **1.5M–2M:** Rapid convergence to near-optimal policy. Eval reward peaks at +6.63, train reward reaches +6.35.

Best episode reward: +7.91. This is close to the theoretical maximum of $3 \times 1.0 + 5.0 - \text{steps} \times 0.01 = +7.0 \sim +7.9$, confirming the agent learned the near-optimal strategy.

6 Discussion

6.1 Why Each Design Change Mattered

Change	Best Reward	Success	What it solved
Baseline (12×12, p=0.5, no shaping)	−3.00	0%	—
+ Reward shaping	−2.69 (train)	0%	Navigation toward landmarks
+ Low penalty (p=0.1)	−2.62 (train)	0%	Made exploration rational
+ Small grid (6×6, K=3)	+6.09	**87.5%**	Spatial coverage & complexity

Table 4: Ablation summary: each design change addresses a specific failure mode.

6.2 The Three Barriers to POMDP Learning

Our experiments reveal three independent barriers that must all be overcome:

1. **Exploration incentive:** The expected reward of a single visit attempt must be positive ($p < 1/K \times r_{\text{correct}}/(1 - 1/K)$). With $K = 4$ and $r_{\text{correct}} = 1.0$, this requires $p < 0.33$. Our original $p = 0.5$ violated this, making "never visit" the rational policy.

2. **Spatial coverage:** The agent's ego-view must cover enough of the grid that landmarks are frequently visible. On a 12×12 grid with 5×5 view (17% coverage), the agent spends most of its time navigating blind. On 6×6 (69% coverage), landmarks are almost always visible.

3. **Memory capacity:** The policy must maintain beliefs over $K!$ orderings. RecurrentPPO (LSTM) succeeds where PPO (MLP) fails, even with the hand-crafted history vector—confirming that **learned memory** is essential for this POMDP.

6.3 Feed-Forward vs. Recurrent: A POMDP Litmus Test

The stark difference between PPO (0% success, all configurations) and RecurrentPPO (87.5% on the feasible configuration) demonstrates that this environment is a **genuine POMDP** that cannot be solved by Markov approximations. The history vector h_t is a sufficient statistic in theory, but the MLP-based PPO cannot learn to use it effectively—the LSTM discovers its own internal representation that better supports sequential decision-making.

6.4 Scaling Considerations

Our results suggest that the 12×12, $K = 4$ configuration is solvable in principle but requires:

- Larger view size (e.g., 7×7 or 9×9) or global position information in the observation.
- Longer training (possibly >50M steps) with curriculum learning.
- Larger LSTM capacity and possibly attention-based architectures.

6.5 Limitations

- We only achieved success on the 6×6 configuration; the 12×12 version remains unsolved.
- The EvalCallback saves models based on unmodified reward, so shaped training + unmodified eval creates a mismatch where "never visit" appears optimal to the callback.
- Only 3 seeds were run; larger seed counts would give tighter confidence intervals.

7 Conclusion

We presented Sequential Multi-Target Delivery, a novel POMDP environment where a robot must infer a hidden visiting order through trial-and-error. Through systematic experimentation, we identified three barriers to learning—exploration incentive, spatial coverage, and memory capacity—and showed that all three must be addressed simultaneously.

Our main findings:

1. **Feed-forward methods fail:** PPO and DQN converge to a "never visit" avoidance strategy (-3.0 reward, 0% success) across all configurations tested.
2. **Reward shaping + penalty tuning are necessary but not sufficient:** They improve training reward but do not achieve task completion on a large grid.
3. **RecurrentPPO succeeds** when the problem is scaled to a feasible difficulty: 6×6 grid, $K = 3$ landmarks, with reward shaping and low wrong-visit penalty. It achieves **87.5% success rate** (seed 0) and near-optimal reward (+6.09, best +7.91).
4. **Seed sensitivity matters:** Out of 3 random seeds, 2 solve the task (87.5% and 78.5% success) while 1 fails entirely (0%). This highlights the role of early exploration luck in RL and motivates future work on exploration robustness.
5. **The iterative debugging process**—from total failure to 87.5% success—demonstrates the importance of systematic ablation in RL research.

Appendix: Hyperparameters

Parameter	PPO	RecurrentPPO	DQN	QR-DQN
Policy network	$[256, 256]$ MLP	$[128, 128]$ MLP + LSTM(128)	$[256, 256]$ MLP	$[256, 256]$ MLP
Learning rate	3×10^{-4}	3×10^{-4}	2.5×10^{-4}	2.5×10^{-4}
Batch size	512	256	128	128
n_{steps} / buffer	512	256	200k	200k
Discount γ	0.99	0.99	0.99	0.99
Parallel envs	16	16	1	1
n_{epochs}	5	5	—	—
Exploration	ent. coef. 0.01	ent. coef. 0.01	ε: $1.0 \to 0.05$	ε: $1.0 \to 0.05$
Total steps	5M	2–5M	5M	5M

Table 5: Hyperparameter summary.

Env Parameter	Large (Phase 1–3)	Small (Phase 4)
Grid size N	12	6
Landmarks K	4	3
Orderings $K!$	24	6
View size	5×5	5×5
View coverage	17%	69%
Max steps T	200	100
Wrong-visit penalty p	0.5	0.1
Reward shaping	varies	Yes

Table 6: Environment configurations across experimental phases.